Write all the required equations (without simplification) to solve the following problems, then find the final answers.

Prob.1: Consider a circuit where a voltage source with $v(t) = 100\cos(1000t + 30°)$ supplies a [illegible] current $i(t) = 10\cos(1000t + 80°)$.

1. The equivalent impedance at the terminals of the voltage source is $Z = 5 \angle 50$

2. The current leads the voltage by 50° degree

3. The voltage leads the current by $\frac{5\pi}{18}$ radian

4. The current value $i(t)$ at $t = 100\mu s$ is 7.83

5. The phasor domain representation of the voltage is $50 \angle 30$

6. The frequency of the current in Hz is 159.154

7. Draw the phasor diagram for both $v(t)$ and $i(t)$

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Prob.2: Consider the circuit shown in Figure 1 use superposition to find $v_0(t)$. Note that the current source is dc

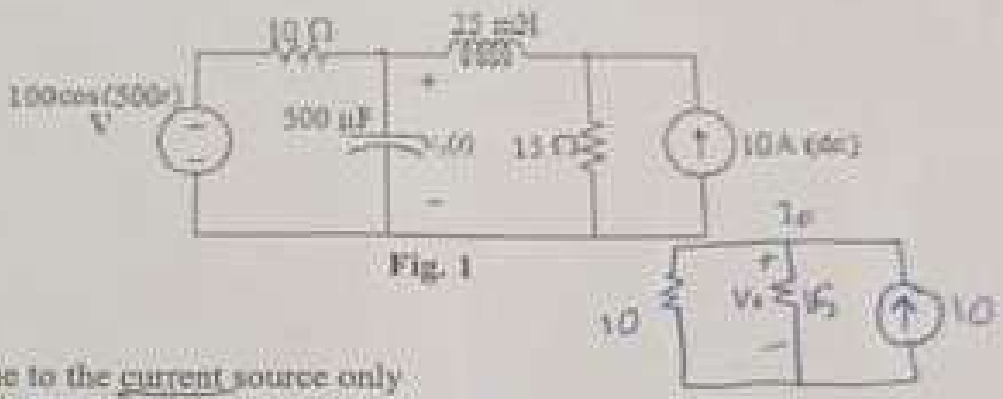


Fig. 1

a. The voltage **V0** due to the current source only

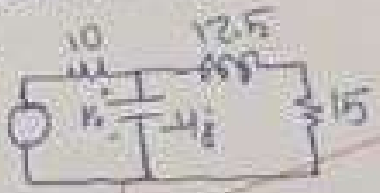
$I_o = \frac{10 \times 10}{10 + 15} = 0.66A$ $V_o = I_o \times R$

$= 0.66 \times 15 = 10$ $V_o = 10\ V$

b. The voltage V_0 due to the voltage source only

$V_o = \frac{100 \angle -45°}{10 - 4j} = 37.139 \angle -18.44°$

$= 13.79 - 34.48j$



$V_0 = 13.79 -$

c. The value of V_0 due to both sources

$V_o = V_o' + V_o''$ $10 + 13.79 - 34j$

$V_0 = 41.82$

$23.79 - 3$

Prob3. Consider the circuit shown in Figure 2.

1. The Thevenin voltage (Vth) with respect to terminals a-b of the circuit.

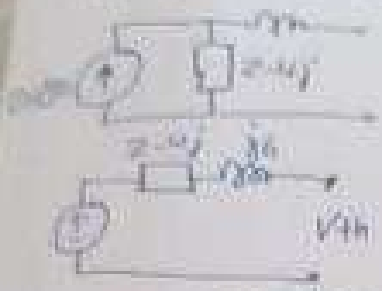




Fig 2



2. The Norton current(I_N) with respect to terminals a-b of the circuit.

3. The Thevenin equivalent impedance between the *a-b* terminals

4. Draw the Thevenin equivalent circuit, then find the load current when $Z_L = 3 - j4\ \Omega$ is connected between the *a-b* terminals

Prob. 4 : ***Consider the circuit shown in Figure 3.***

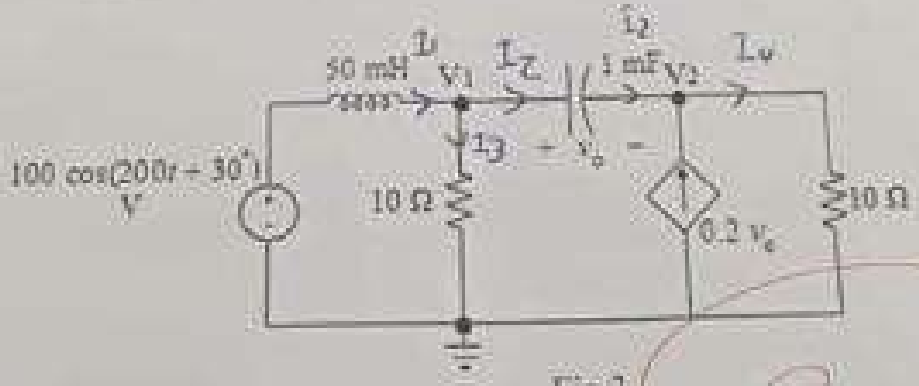


Fig 3

Using the **nodal method**, write the required equations that you are needed to find V_1, V_2, and V_0

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Good luck